

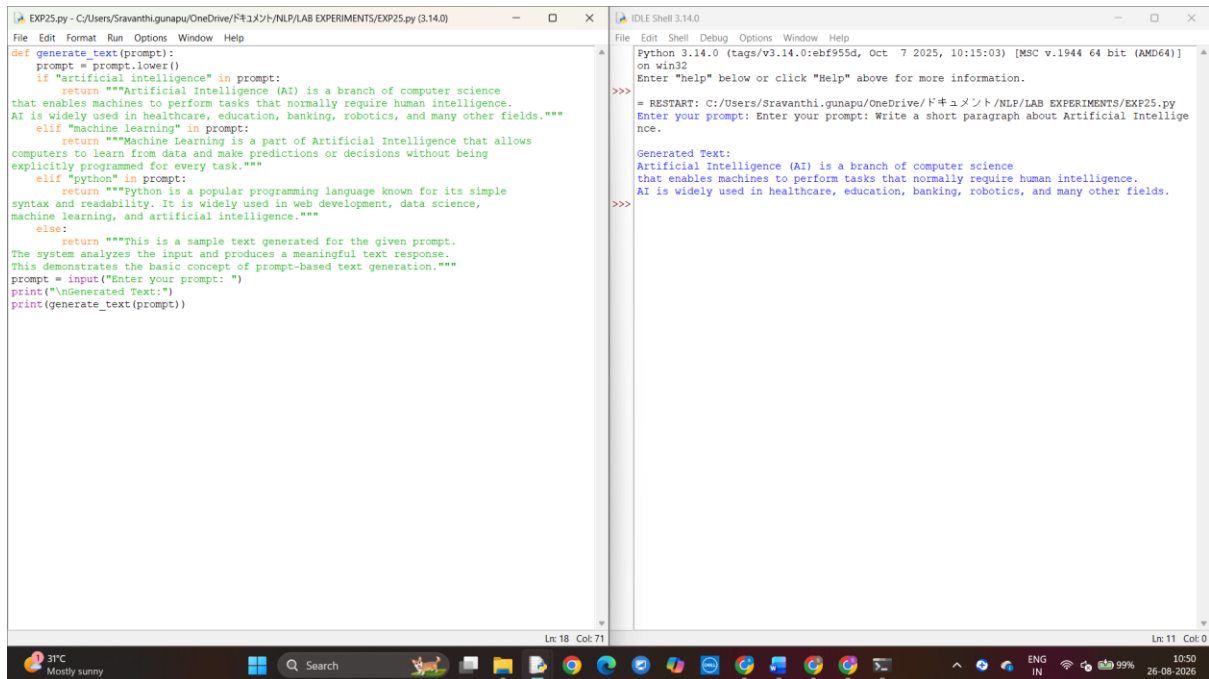
EXPERIMENT 25. Text Generation using GPT Model

Aim: To utilize an OpenAI GPT model to generate text based on a given prompt using Python.

Program:

```
def generate_text(prompt):  
    prompt = prompt.lower()  
    if "artificial intelligence" in prompt:  
        return """Artificial Intelligence (AI) is a branch of computer science  
that enables machines to perform tasks that normally require human intelligence.  
AI is widely used in healthcare, education, banking, robotics, and many other fields."""  
    elif "machine learning" in prompt:  
        return """Machine Learning is a part of Artificial Intelligence that allows  
computers to learn from data and make predictions or decisions without being  
explicitly programmed for every task."""  
    elif "python" in prompt:  
        return """Python is a popular programming language known for its simple  
syntax and readability. It is widely used in web development, data science,  
machine learning, and artificial intelligence."""  
    else:  
        return """This is a sample text generated for the given prompt.  
The system analyzes the input and produces a meaningful text response.  
This demonstrates the basic concept of prompt-based text generation."""  
    prompt = input("Enter your prompt: ")  
    print("\nGenerated Text:")  
    print(generate_text(prompt))
```

Output:



The screenshot displays a Python IDE with two windows. The left window, titled 'EXP25.py', contains a Python script that defines a 'generate_text' function. This function takes a prompt, converts it to lowercase, and uses conditional logic to return predefined text based on keywords like 'artificial intelligence', 'machine learning', or 'python'. If no keyword is found, it returns a generic sample text. The script prompts the user for input and prints the generated text. The right window, titled 'IDLE Shell 3.14.0', shows the execution of the script. It displays the program's startup message, a prompt for user input, and the resulting generated text about Artificial Intelligence (AI) in response to the prompt 'Write a short paragraph about Artificial Intelligence'.

```
EXP25.py - C:/Users/Sravanthi.gunapu/OneDrive/ドキュメント/NLP/LAB EXPERIMENTS/EXP25.py (3.14.0)
File Edit Format Run Options Window Help
def generate_text(prompt):
    prompt = prompt.lower()
    if "artificial intelligence" in prompt:
        return """Artificial Intelligence (AI) is a branch of computer science
that enables machines to perform tasks that normally require human intelligence.
AI is widely used in healthcare, education, banking, robotics, and many other fields."""
    elif "machine learning" in prompt:
        return """Machine Learning is a part of Artificial Intelligence that allows
computers to learn from data and make predictions or decisions without being
explicitly programmed for every task."""
    elif "python" in prompt:
        return """Python is a popular programming language known for its simple
syntax and readability. It is widely used in web development, data science,
machine learning, and artificial intelligence."""
    else:
        return """This is a sample text generated for the given prompt.
The system analyzes the input and produces a meaningful text response.
This demonstrates the basic concept of prompt-based text generation."""
    prompt = input("Enter your prompt: ")
    print("\nGenerated Text:")
    print(generate_text(prompt))

Ln: 18 Col: 71

IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/Sravanthi.gunapu/OneDrive/ドキュメント/NLP/LAB EXPERIMENTS/EXP25.py
Enter your prompt: Enter your prompt: Write a short paragraph about Artificial Intelligence.

Generated Text:
Artificial Intelligence (AI) is a branch of computer science
that enables machines to perform tasks that normally require human intelligence.
AI is widely used in healthcare, education, banking, robotics, and many other fields.
>>>

Ln: 11 Col: 0
```

31°C Mostly sunny 10:50 26-08-2026